Supplementary appendix

Factor-combination discovery is stochastic threshold group testing: a counterfactual replay of the 2006 reprogramming screen

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This appendix carries the campaigns behind the intervals and paired tests in the main text, the figures the page limit displaced, and the complete statistical report as the analysis code emits it. Every number in the main text can be traced to a section here, and every section here can be regenerated from the public repository (`https://github.com/mashal-alkhalifah/yamanaka-replay`) without rerunning any simulation: the per-run CSVs are committed, and `analysis/analyze.py` reads them.

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S1 The experimental campaign

Reference environment: Linux 6.18, x86_64, 2 cores, 7.8 GB RAM, Python 3.13.16, NumPy 2.5.3. No GPU and no network access are required. The statistics layer is implemented directly in NumPy to hold the dependency list at one package, and is therefore tested: `analysis/test_stats.py` checks each estimator against a closed-form or planted-truth answer, including the permutation test's false-positive rate under the null.

Three corrections this campaign forced on its own earlier claims. An earlier version of this work reported that absent graded synergy was a structural failure, that the design step was independent of hypothesis-space size, and that the adaptive designer's advantage ended at 48 candidates. Experiment 6 refuted the first (Section S2), experiment 8 refuted the second, and experiment 15 refuted the third (main text §3.2). They are recorded here rather than quietly dropped, because a reader deciding how much to trust the remaining claims should know how the earlier ones fared.

Script	What it varies	Runs	Supports
`01_main_comparison`	the four strategies, small	80	demonstration
`02_prior_robustness`	prior quality, small	60	demonstration
`03_misspecification`	one assumption at a time	120	S2
`04_trace_single_run`	one run, experiment by experiment	1	demonstration
`05_scale_main`	the four strategies, paired	1,200	main text §3.1
`06_synergy_threshold`	true vs assumed synergy	1,100	S2
`07_stopping_rule`	stopping threshold	1,680	S3
`08_scaling`	24 to 60 candidates	300	main text §3.2
`09_prior_quality`	strength × coverage × breadth	1,800	S4
`10_unknown_cardinality`	fixed vs latent recipe size	240	main text §3.3
`11_classical_baseline`	half-removal across sizes	270	main text §3.2
`12_cardinality_baselines`	two-stage and model-free	225	main text §3.3
`13_measured_gap`	ρ drawn from measurement	2,200	main text §3.4
`14_norman_retrospective`	measured queries only	120	main text §3.6
`15_proposal_ablation`	only the proposal set	184	main text §3.2
`16_latent_gap`	inferring ρ vs a pilot	1,500	main text §3.5

Table S1: Every campaign, with the claim it supports. Runs are independent replicates recorded individually, so all aggregation is reproducible without re-simulation. Strategies within an experiment share hidden recipes and noise seeds, making every comparison paired.

S2 Model error: absent signal versus wrong signal

Making the oracle disagree with the designer one assumption at a time (experiment 3) suggested that removing graded synergy was fatal: 43% success. That experiment conflated two things, removing the partial signal from the assay while leaving the searcher believing in it. Experiment 6 separates them. With no graded synergy whatsoever and a searcher that knows it, the method succeeds in 100% of runs at a median of 22 experiments: it degenerates into pure group testing, pays 2.2× for the lost gradient, and is still 64% cheaper than the 2006 protocol. The earlier figure was misspecification, not lost information.

What does matter is how wrong the assumed value is and in which direction, and the window is asymmetric. Under-estimating the partial effect fivefold still succeeds in 70% of runs; over-estimating it fourfold drops to 42%, and reports a confidence of 0.989 while doing so. This asymmetry is what makes the measured audit in the main text consequential: the calibrated 0.10 sits on the dangerous side of the true value, not the safe one.

S3 The stopping rule

Raising the stopping threshold from 0.50 to 0.99 costs little and buys reliability, so the rule is not a tuning knob the results depend on. Under a correctly specified model the reported confidence tracks the realised success rate closely. That calibration is conditional on the model, and the main text shows what happens when it is not: the same rule returns 0.991 on runs that end wrong. A stopping rule can only be as honest as the likelihood behind it.

S4 Prior quality

A literature prior up-weighting eight factors reduces the median from 10 to 6 experiments. Experiment 9 varies its quality along three axes — strength, coverage (how many true factors it includes) and breadth (how many decoys) — over 1,800 runs. The prior degrades gracefully: a factor the literature missed still carries non-zero mass and can be recovered from data, so the method does not fail when the prior is wrong, it merely stops helping. The one reading to avoid is treating the prior's benefit as scale-free. In experiment 8 the informed arm appears flat in n only because prior quality is held constant, up-weighting eight factors

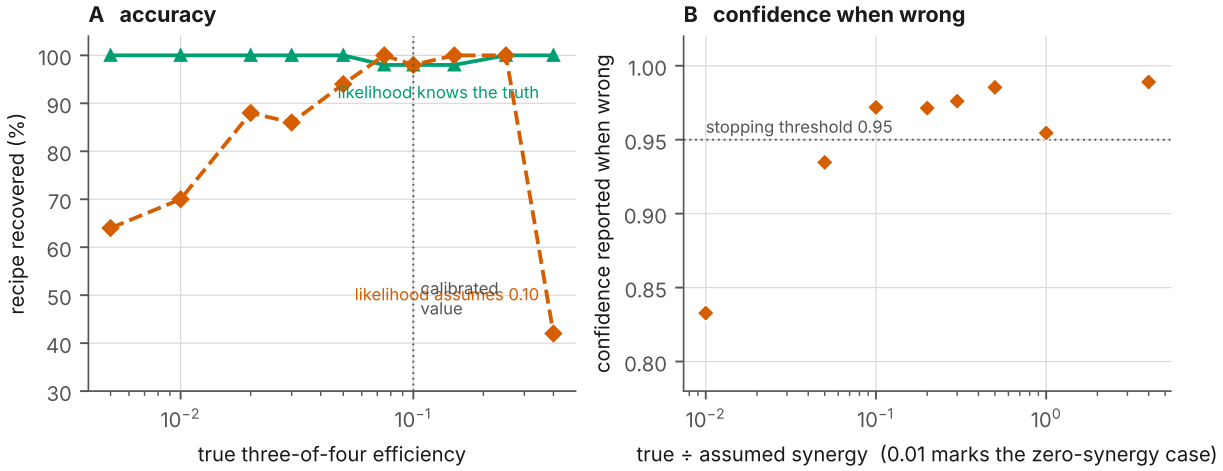


Figure S1: True three-of-four efficiency against what the searcher assumes. Arms are paired by run, so only the searcher's assumption differs. A searcher told the truth succeeds at every true value tested, including zero; a searcher holding the calibrated 0.10 degrades asymmetrically, and degrades worst when it *over*-estimates the partial effect.

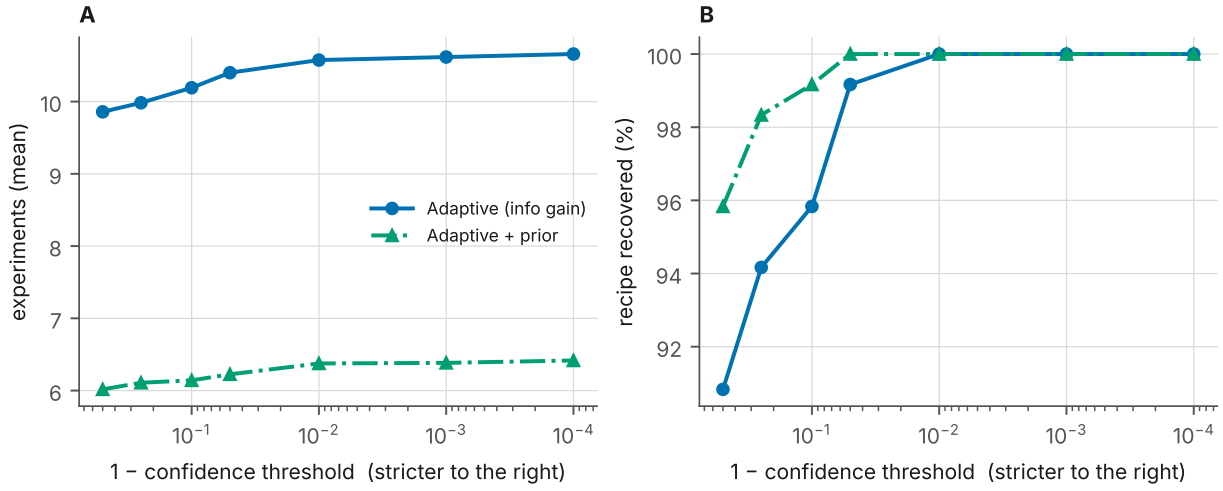


Figure S2: Cost and reliability against the stopping threshold. The rule is nearly free over the usable range, and the posterior is well calibrated *when the model is right* — which is the condition the main text's §3.4 removes.

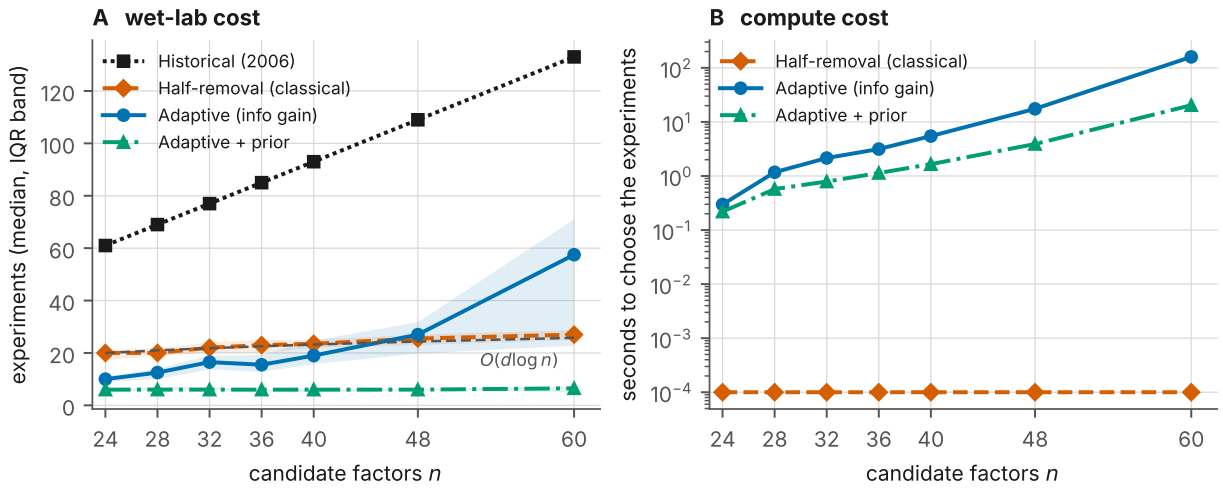


Figure S3: Growth with candidate-list size, as first measured. (A) Wet-lab cost, medians with interquartile bands, against the $O(d \log n)$ reference curve scaled at the smallest list. (B) Compute spent choosing experiments. This is the measurement that produced the regime claim the main text then withdraws.

regardless of list length; a longer candidate list in practice implies a less informative prior, so that curve is an upper bound conditional on prior quality, not a prediction.

S5 Scaling, the classical baseline, and the proposal generator

Compute cost grows as roughly $H^{1.5}$ in the hypothesis count H. The correct narrow claim is that the design step is independent of *how concentrated the posterior is* — the $k+1$ overlap-class collapse is real — but it remains linear in the *size of the enumerated space*, and the number of rounds grows too. Exact enumeration is the binding constraint past about 60 candidates, and the structured proposals of the main text do not change this; they change only the number of wet-lab experiments.

The structured generator is also faster in wall-clock terms despite proposing larger pools, because it needs fewer rounds: 89 s per run at 60 candidates against 173 s. Part of that is a separate exact optimisation, counting overlaps over whichever side of the pool is smaller, verified against the direct computation on 400 random pools at two candidate counts.

S6 The measured audit in full

Two observations beyond the headline median. First, departures from additivity are common but modest: in Norman the median ratio of a pair's effect to the sum of its singles is 0.91, with 47% sub-additive and 27% super-additive, and only 28% of pairs have an additivity residual exceeding their own sampling noise, estimated by regressing squared residuals on $1/n_{AB} + 1/n_A + 1/n_B$. Second, the two systems differ in *where* the interaction lives: the median R^2 of a pair on its two singles is 0.86 in K562 but 0.64 in hESCs, so combinations in the stem-cell system leave the plane spanned by their constituents more while their magnitudes stay near-additive.

The result is not an artefact of measuring effect size by a Euclidean norm. Repeating the Norman analysis with threshold-like functionals — the number of genes moving more than 0.25 or 0.5 in log space — lowers the median to 0.48 and 0.50 and raises the fraction below 0.20 to 25% and 22%, but no functional produces a median near 0.10.

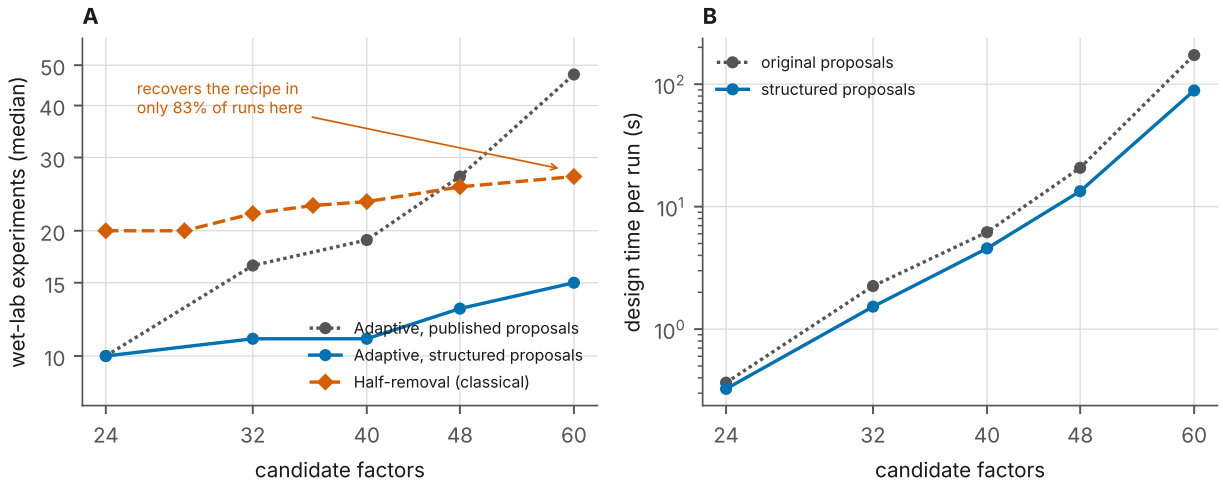


Figure S4: Changing only which pools may be scored. (A) Median wet-lab cost against candidate-list size for the published generator, the structured one, and half-removal. Scoring rule, posterior, oracle, recipes, seeds and the number of pools scored per round are identical across the two adaptive curves. (B) The design-time price of the change, which is negative: the structured proposals need fewer rounds than their larger pools cost.

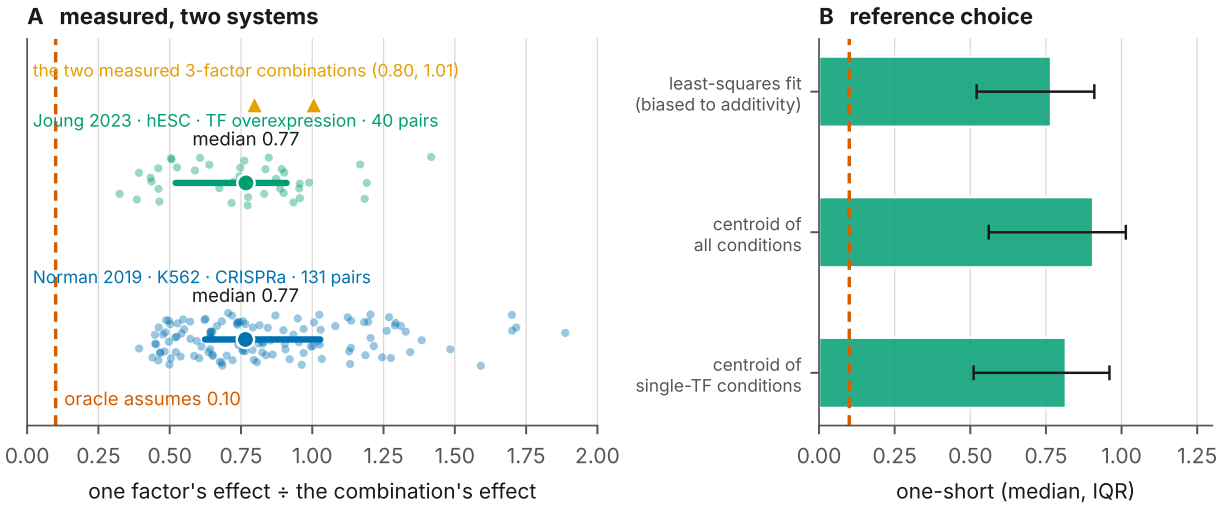


Figure S5: What one factor of a measured combination reproduces. (A) The one-short ratio for 131 gene pairs in K562 [1] and 40 transcription-factor pairs in human embryonic stem cells [2], with the two measured three-factor combinations marked. (B) The hESC subset has no unperturbed control, so three references are reported; the conclusion holds under all three.

S7 Complete statistical report

What follows is the verbatim output of `analysis/analyze.py`, which reads the committed per-run CSVs and runs no simulation. Box-drawing characters have been transliterated to ASCII for typesetting; the file itself is in the repository at `results/analysis_report.txt`.

```
================================================================================
  Experiment 5 -- main comparison, 300 paired replicates
================================================================================
  strategy                          median     95% CI   mean      IQR   p90   max   success (95% CI)
  ------------------------------------------------------------------------------------------------------
  Historical protocol (2006)           61    [61,61]   61.0    61-61    61    61   100.0% [98.7%, 100.0%]
  Adaptive design (info gain)          10    [10,10]   10.3     9-11    13    18   99.7% [98.1%, 99.9%]
  Adaptive + literature prior           6      [6,6]    6.2      6-7     7    11   99.3% [97.6%, 99.8%]
  Random pools + Bayes [ablation]      44    [42,46]   43.0    33-56    60    60   88.0% [83.8%, 91.2%]

  Saving vs the 2006 protocol (median 61 experiments):
    Adaptive design (info gain)         84%   (6.1x fewer experiments)
    Adaptive + literature prior         90%   (10.2x fewer experiments)
    Random pools + Bayes [ablation]     28%   (1.4x fewer experiments)

================================================================================
  Paired tests -- same hidden recipe, same noise seed, different strategy
================================================================================

  Adaptive design (info gain) vs Random pools + Bayes [ablation]
    is informative pool choice doing the work?
    mean difference -32.7 experiments   permutation p = < 1e-4
    cheaper in 298/300 runs, tied in 1
    accuracy: adaptive only 36, random only 1   McNemar p = < 1e-4

  Adaptive + literature prior vs Adaptive design (info gain)
    does the literature prior add anything?
    mean difference -4.1 experiments   permutation p = < 1e-4
    cheaper in 294/300 runs, tied in 3
    accuracy: informed only 1, adaptive only 2   McNemar p = 1.0000

  Adaptive design (info gain) vs Historical protocol (2006)
    vs the historical protocol
    mean difference -50.7 experiments   permutation p = < 1e-4
    cheaper in 300/300 runs, tied in 0
    accuracy: adaptive only 0, historical only 1   McNemar p = 1.0000

================================================================================
  Tail behaviour -- the number a lab would actually plan against
================================================================================
  Adaptive design (info gain)     worst run  18   P(>=61, i.e. no better than 2006) = 0.0%   failures 1
      failing runs used 8 experiments (median) vs 10 for successes
  Adaptive + literature prior     worst run  11   P(>=61, i.e. no better than 2006) = 0.0%   failures 2
      failing runs used 6 experiments (median) vs 6 for successes
  Random pools + Bayes [ablation] worst run  60   P(>=61, i.e. no better than 2006) = 0.0%   failures 36
      failing runs used 60 experiments (median) vs 42 for successes

================================================================================
  Is failure detectable from the inside?
================================================================================
  Adaptive design (info gain)     mean reported confidence: 0.995 when right, 0.955 when wrong
  Adaptive + literature prior     mean reported confidence: 0.996 when right, 0.968 when wrong
  Random pools + Bayes [ablation] mean reported confidence: 0.956 when right, 0.423 when wrong

================================================================================
  Experiment 6 -- how much graded synergy does the method need?
================================================================================
  three-of-four efficiency, and what the searcher believes about it

   partial |  matched: median   success |  misspec: median   success
  ------------------------------------------------------------------------
     0.000 |               22    100.0% |               35     52.0% |
     0.005 |               18    100.0% |               23     64.0% |
     0.010 |               19    100.0% |               18     70.0% |
     0.020 |               14    100.0% |               15     88.0% |
     0.030 |               13    100.0% |               13     86.0% |
     0.050 |               12    100.0% |               12     94.0% |
     0.075 |               11     98.0% |               10    100.0% |
     0.100 |               10     98.0% |               10     98.0% |
     0.150 |               10     98.0% |               10    100.0% |
     0.250 |               10    100.0% |               10    100.0% |
     0.400 |               10    100.0% |               12     42.0% |

  Expected colonies from a three-of-four pool, for scale:
    partial 0.10 -> ~4 colonies (the calibrated value, i.e. the 2006 assay)
    partial 0.02 -> ~1 colony     partial 0.005 -> ~0.2 colonies

================================================================================
  Where the method breaks, and why
================================================================================
  matched
    lowest success anywhere in the sweep: 98.0%
    no 50% crossing inside the tested range
    no 90% crossing inside the tested range
  misspecified
    lowest success anywhere in the sweep: 42.0%
    no 50% crossing inside the tested range
    no 90% crossing inside the tested range

================================================================================
  The real axis is not synergy strength -- it is calibration error
================================================================================
  The misspecified searcher always assumes partial = 0.10. Success as a
  function of how wrong that assumption is:

   true/assumed | true partial |  success |  median |  reported confidence when wrong
```

```
  ------------------------------------------------------------------------------
            0x |        0.000 |    52.0% |      35 |                          0.833
         0.05x |        0.005 |    64.0% |      23 |                          0.935
         0.10x |        0.010 |    70.0% |      18 |                          0.972
         0.20x |        0.020 |    88.0% |      15 |                          0.971
         0.30x |        0.030 |    86.0% |      13 |                          0.976
         0.50x |        0.050 |    94.0% |      12 |                          0.985
         0.75x |        0.075 |   100.0% |      10 |                             --
         1.00x |        0.100 |    98.0% |      10 |                          0.955
         1.50x |        0.150 |   100.0% |      10 |                             --
         2.50x |        0.250 |   100.0% |      10 |                             --
         4.00x |        0.400 |    42.0% |      12 |                          0.989

================================================================================
  How much of the failure is a modelling error rather than lost information?
================================================================================
  Paired by run: same hidden recipe and noise, differing only in whether
  the searcher's likelihood knows the true partial efficiency.

    partial  0.005:  matched 100.0%   misspecified  64.0%   gap   +36 pts   McNemar p = < 1e-4  <-- calibration matters here
    partial  0.010:  matched 100.0%   misspecified  70.0%   gap   +30 pts   McNemar p = < 1e-4  <-- calibration matters here
    partial  0.020:  matched 100.0%   misspecified  88.0%   gap   +12 pts   McNemar p = 0.0313  <-- calibration matters here
    partial  0.030:  matched 100.0%   misspecified  86.0%   gap   +14 pts   McNemar p = 0.0156  <-- calibration matters here
    partial  0.050:  matched 100.0%   misspecified  94.0%   gap    +6 pts   McNemar p = 0.2500
    partial  0.075:  matched  98.0%   misspecified 100.0%   gap    -2 pts   McNemar p = 1.0000
    partial  0.100:  matched  98.0%   misspecified  98.0%   gap    +0 pts   McNemar p = 1.0000
    partial  0.150:  matched  98.0%   misspecified 100.0%   gap    -2 pts   McNemar p = 1.0000
    partial  0.250:  matched 100.0%   misspecified 100.0%   gap    +0 pts   McNemar p = 1.0000
    partial  0.400:  matched 100.0%   misspecified  42.0%   gap   +58 pts   McNemar p = < 1e-4  <-- calibration matters here

================================================================================
  Experiment 7 -- the price of certainty, and is the posterior honest?
================================================================================

  Adaptive design (info gain)
    target |  median    mean    p90 |   success           95% CI |  mean reported     gap
  ---------------------------------------------------------------------------------------
    0.5000 |      10     9.9     12 |     90.8%   [84.3%, 94.8%] |           0.924  +0.015
    0.7500 |      10    10.0     12 |     94.2%   [88.4%, 97.1%] |           0.952  +0.010
    0.9000 |      10    10.2     13 |     95.8%   [90.6%, 98.2%] |           0.983  +0.025
    0.9500 |      10    10.4     13 |     99.2%   [95.4%, 99.9%] |           0.995  +0.003
    0.9900 |      10    10.6     13 |    100.0%  [96.9%, 100.0%] |           1.000  -0.000
    0.9990 |      10    10.6     13 |    100.0%  [96.9%, 100.0%] |           1.000  -0.000
    0.9999 |      10    10.7     13 |    100.0%  [96.9%, 100.0%] |           1.000  -0.000

  Adaptive + literature prior
    target |  median    mean    p90 |   success           95% CI |  mean reported     gap
  ---------------------------------------------------------------------------------------
    0.5000 |       6     6.0      7 |     95.8%   [90.6%, 98.2%] |           0.956  -0.002
    0.7500 |       6     6.1      7 |     98.3%   [94.1%, 99.5%] |           0.986  +0.003
    0.9000 |       6     6.1      7 |     99.2%   [95.4%, 99.9%] |           0.991  -0.000
    0.9500 |       6     6.2      7 |    100.0%  [96.9%, 100.0%] |           0.996  -0.004
    0.9900 |       6     6.4      7 |    100.0%  [96.9%, 100.0%] |           1.000  -0.000
    0.9990 |       6     6.4      8 |    100.0%  [96.9%, 100.0%] |           1.000  -0.000
    0.9999 |       6     6.4      8 |    100.0%  [96.9%, 100.0%] |           1.000  -0.000

================================================================================
  Calibration verdict
================================================================================
  Adaptive design (info gain): no threshold where the stated confidence exceeds the observed success rate
  Adaptive + literature prior: no threshold where the stated confidence exceeds the observed success rate

================================================================================
  Marginal cost of certainty
================================================================================
  Adaptive design (info gain): 0.5000->0.7500: +0   0.7500->0.9000: +0   0.9000->0.9500: +0   0.9500->0.9900: +0   0.9900->0.9990: +0   0.9990->0.9999: +0
  Adaptive + literature prior: 0.5000->0.7500: +0   0.7500->0.9000: +0   0.9000->0.9500: +0   0.9500->0.9900: +0   0.9900->0.9990: +0   0.9990->0.9999: +0

================================================================================
  Experiment 8 -- scaling in candidate-list size
================================================================================
   cands   hypotheses |  historical |  adaptive  success  sec/run |  informed  success
  ------------------------------------------------------------------------------
      24       10,626 |          61 |        10   100.0%      0.3 |         6   100.0%
      32       35,960 |          77 |        16   100.0%      2.1 |         6   100.0%
      40       91,390 |          93 |        19   100.0%      5.5 |         6   100.0%
      48      194,580 |         109 |        27   100.0%     17.4 |         6   100.0%
      60      487,635 |         133 |        58   100.0%    159.2 |         6   100.0%

================================================================================
  Growth rates
================================================================================
  Historical protocol (2006)       12.93 experiments per bit of hypothesis-space entropy
  Adaptive design (info gain)       7.63 experiments per bit of hypothesis-space entropy
  Adaptive + literature prior       0.07 experiments per bit of hypothesis-space entropy

  wet-lab cost vs candidates : exponent 1.77   (1.00 would be linear, as the historical protocol is)
  compute cost vs hypotheses : exponent 1.56   (design-step cost, seconds per run)

  Saving against the historical protocol at each size:
     24 candidates: adaptive  84%   informed  90%
     32 candidates: adaptive  79%   informed  92%
     40 candidates: adaptive  80%   informed  94%
     48 candidates: adaptive  75%   informed  94%
     60 candidates: adaptive  57%   informed  95%

================================================================================
  Experiment 9 -- when does a literature prior stop helping and start hurting?
================================================================================

  Prior up-weights 6 factors (2 of them decoys when coverage is complete)
   strength | drop 0:  med  succ | drop 1:  med  succ | drop 2:  med  succ | drop 3:  med  succ | drop 4:  med  succ
  ---------------------------------------------------------------------------------------------------------------
        1.5 |          7 100.0% |          8 100.0% |          9 100.0% |         10 100.0% |         11 100.0%
        3.0 |          6 100.0% |          7 100.0% |         10 100.0% |         11 100.0% |         12 100.0%
       10.0 |          5 100.0% |          7 100.0% |          9 96.7% |         12 100.0% |         14 93.3%
      100.0 |          4 100.0% |          7 100.0% |         11 96.7% |         15 96.7% |         18 90.0%
```

```
  Prior up-weights 8 factors (4 of them decoys when coverage is complete)
   strength | drop 0:  med  succ | drop 1:  med  succ | drop 2:  med  succ | drop 3:  med  succ | drop 4:  med  succ
  ------------------------------------------------------------------------------------------------------------
        1.5 |          7 100.0% |          9 100.0% |          9 100.0% |         11 93.3% |         12 100.0%
        3.0 |          6 100.0% |          8 100.0% |         10 100.0% |         12 100.0% |         13 100.0%
       10.0 |          5 100.0% |          8 100.0% |         10 100.0% |         13 96.7% |         15 100.0%
      100.0 |          5 96.7% |          9 100.0% |         12 100.0% |         17 90.0% |         21 86.7%

  Prior up-weights 12 factors (8 of them decoys when coverage is complete)
   strength | drop 0:  med  succ | drop 1:  med  succ | drop 2:  med  succ | drop 3:  med  succ | drop 4:  med  succ
  ------------------------------------------------------------------------------------------------------------
        1.5 |          8 100.0% |         10 100.0% |         10 100.0% |         11 100.0% |         12 100.0%
        3.0 |          8 100.0% |          9 100.0% |         11 96.7% |         12 100.0% |         14 100.0%
       10.0 |          7 96.7% |          9 100.0% |         13 100.0% |         15 100.0% |         17 96.7%
      100.0 |          7 100.0% |         11 96.7% |         16 90.0% |         20 96.7% |         24 96.7%

================================================================================
  The trade: a strong prior is cheap when right and dangerous when wrong
================================================================================
   strength |  median when complete |  median when 1 missing |  success when 1 missing
  ------------------------------------------------------------------------------------
        1.5 |                     8 |                     9 | 100.0% [95.9%, 100.0%]
        3.0 |                     6 |                     8 | 100.0% [95.9%, 100.0%]
       10.0 |                     6 |                     8 | 100.0% [95.9%, 100.0%]
      100.0 |                     5 |                     9 | 98.9% [94.0%, 99.8%]

  Failure rate by coverage, pooled over breadth, per strength:
    strength    1.5   drop 0: 0.0%   drop 1: 0.0%   drop 2: 0.0%   drop 3: 2.2%   drop 4: 0.0%
    strength    3.0   drop 0: 0.0%   drop 1: 0.0%   drop 2: 1.1%   drop 3: 0.0%   drop 4: 0.0%
    strength   10.0   drop 0: 1.1%   drop 1: 0.0%   drop 2: 1.1%   drop 3: 1.1%   drop 4: 3.3%
    strength  100.0   drop 0: 1.1%   drop 1: 1.1%   drop 2: 4.4%   drop 3: 5.6%   drop 4: 8.9%

================================================================================
  Experiment 10 -- recipe size as a latent variable instead of an assumption
================================================================================
   true size |  designer |  median    mean |  exact recipe            95% CI |  right size |  P(true size)
  ------------------------------------------------------------------------------------------------------
           3 |   fixed-4 |       9     8.7 |          0.0%      [0.0%, 8.8%] |        0.0% |         0.000
           3 |    latent |       9     8.9 |        100.0%   [91.2%, 100.0%] |      100.0% |         0.999
           4 |   fixed-4 |      10    10.5 |        100.0%   [91.2%, 100.0%] |      100.0% |         1.000
           4 |    latent |      12    11.4 |        100.0%   [91.2%, 100.0%] |      100.0% |         0.997
           5 |   fixed-4 |      18    18.9 |          0.0%      [0.0%, 8.8%] |        0.0% |         0.000
           5 |    latent |      16    16.3 |        100.0%   [91.2%, 100.0%] |      100.0% |         0.995

================================================================================
  Does the fix cost anything on the case it did not need to generalise for?
================================================================================
  Four-factor truth, the case the fixed designer was built for:
    fixed-4  median 10, success 100.0%
    latent   median 12, success 100.0%
    premium  +0.9 experiments   permutation p = 0.0036

================================================================================
  Is the failure still silent?
================================================================================
  The danger was never the failure itself but that the posterior stayed
  confident through it. Reported confidence, split by whether the answer
  was right:

    size 3, fixed-4  confidence -- when right, 0.999 when wrong
    size 3, latent   confidence 0.999 when right, -- when wrong
    size 4, fixed-4  confidence 0.997 when right, -- when wrong
    size 4, latent   confidence 0.996 when right, -- when wrong
    size 5, fixed-4  confidence -- when right, 0.987 when wrong
    size 5, latent   confidence 0.993 when right, -- when wrong

================================================================================
  Experiment 12 -- is joint inference over recipe size worth its machinery?
================================================================================
  method                  true size 3           true size 4           true size 5
  ------------------------------------------------------------------------------
  fixed-4                  9 exp . 0.0%       10 exp . 100.0%         18 exp . 0.0%
  two-stage-6             9 exp . 80.0%        10 exp . 46.7%        17 exp . 13.3%
  two-stage-12         12 exp . 100.0%        12 exp . 93.3%        17 exp . 60.0%
  latent                   9 exp . 100.0%       11 exp . 100.0%       17 exp . 100.0%
  half-removal            17 exp . 100.0%       20 exp . 100.0%       22 exp . 100.0%

================================================================================
  What each comparator costs on the four-factor case it was built for
================================================================================
  two-stage-6      -0.5 experiments vs fixed-4   p = 0.5184
  two-stage-12     +1.7 experiments vs fixed-4   p = 0.0053
  latent           +0.5 experiments vs fixed-4   p = 0.5252
  half-removal     +9.7 experiments vs fixed-4   p = < 1e-4

================================================================================
  Where the two-stage baseline loses
================================================================================
  It commits to a size estimated from a fixed first-stage budget, and
  cannot revisit that decision. Share of runs whose committed size was
  wrong:

    two-stage-6   true size 3: committed wrongly in 20.0% of runs
    two-stage-6   true size 4: committed wrongly in 53.3% of runs
    two-stage-6   true size 5: committed wrongly in 86.7% of runs
    two-stage-12  true size 3: committed wrongly in 0.0% of runs
    two-stage-12  true size 4: committed wrongly in 6.7% of runs
    two-stage-12  true size 5: committed wrongly in 40.0% of runs

================================================================================
  Experiment 13 -- the designer under measured gap structure
================================================================================
  rho drawn from 200 Norman ratios: median 0.77, IQR 0.62-1.02
  the published oracle assumes 0.10

  fall-off shape: gap
```

```
  method                              median    mean             success   silent
  Adaptive at the assumed rho=0.10        10    10.4      99.0% [96-100]      0.0%
  Adaptive, published calibration          9    10.4         2.5% [1-6]     97.5%
  Adaptive, told the true rho             12    11.8      99.0% [96-100]      1.0%
  Half-removal, assumed threshold         16    15.6         0.0% [0-2]      0.0%
  Half-removal, matched threshold         20    17.5         8.5% [5-13]     0.0%
  Historical protocol (2006)              61    61.0       28.5% [23-35]     0.0%

fall-off shape: geometric
  method                              median    mean             success   silent
  Adaptive, published calibration          5     6.1         0.0% [0-2]    100.0%
  Adaptive, told the true rho             16    35.4       88.0% [83-92]      1.5%
  Half-removal, assumed threshold          7    10.7         0.0% [0-2]      0.0%
  Half-removal, matched threshold         16    17.4         7.0% [4-11]     0.0%
  Historical protocol (2006)              61    61.0       28.0% [22-35]     0.0%

gap: published calibration 2.5% vs matched 99.0% success, exact McNemar < 1e-4
geometric: published calibration 0.0% vs matched 88.0% success, exact McNemar < 1e-4

A silent failure is not a hedged one: under the published calibration the
mean final confidence on runs that ended wrong was 0.991.

================================================================================
  Experiment 14 -- retrospective discovery on measured data
================================================================================
  Every query returns a measured Norman condition; nothing is simulated.

  arm               found best  median tests   at 20 tests   at 60 tests
  random                 13/40            84         74.4%         86.1%
  singles-first          40/40           108         59.3%         86.1%
  surrogate              40/40            68         83.3%         92.1%
  exhaustive               1/1           236

At a 20-experiment budget the surrogate reaches 83% of the best measured combination against
59% for the singles-first protocol: difference 20 points, < 1e-4.
Read this as a statement about the landscape as much as the method:
the measured one-short ratio of 0.77 is exactly what makes a singles-first
protocol nearly sufficient here, and what makes the simulated assay
(where a single factor yields nothing) the easier problem of the two.

================================================================================
  Experiment 15 -- proposal generator ablation
================================================================================
  Scoring rule, posterior, oracle, recipes and seeds all held fixed;
  only the set of pools the designer may score changes.

    cands|  original  success   s/run|  structured  success   s/run|  half-removal  success
   ------------------------------------------------------------------------------------------
       24|        10   100.0%     0.4|          10   100.0%     0.3|            20   100.0%
       32|        16   100.0%     2.2|          11   100.0%     1.5|            22    96.7%
       40|        19   100.0%     6.2|          11   100.0%     4.6|            24    96.7%
       48|        27   100.0%    20.8|          13   100.0%    13.4|            26    93.3%
       60|        48   100.0%   172.9|          15   100.0%    88.8|            27    83.3%

================================================================================
  Growth rates (exponent of median cost against candidate count)
================================================================================
  adaptive, original      1.61
  adaptive, structured    0.43
  half-removal (exp 11)     0.36

  Paired comparison at each size (same recipe, same seed):
     24 candidates: structured saves    0.6 experiments on average, 0.2920
     32 candidates: structured saves    5.5 experiments on average, 0.0001
     40 candidates: structured saves    8.6 experiments on average, < 1e-4
     48 candidates: structured saves   19.7 experiments on average, < 1e-4
     60 candidates: structured saves   34.2 experiments on average, 0.0006

  Structured proposals are at or below half-removal's cost at every size tested.

================================================================================
  Experiment 16 -- the gap response as a latent variable
================================================================================
  All arms use structured proposals, the same drawn rho and the same
  recipes, so neither the proposal generator nor the landscape differs.

  true fall-off: gap
    method                          median   mean            success  silent |rho err|
    Fixed at the assumed 0.10           10   11.3         2.7% [1-7]    97.3%      0.67
    Fixed, told the true rho            12   11.5     99.3% [96-100]     0.7%      0.00
    Pilot (5 tests), then fixed         17   16.7      95.3% [91-98]     4.7%      0.09
    Latent rho                          12   12.0     99.3% [96-100]     0.7%      0.06
    Latent rho and shape                12   12.2    100.0% [98-100]     0.0%      0.06

  true fall-off: geometric
    method                          median   mean            success  silent |rho err|
    Fixed at the assumed 0.10            7    8.1         0.0% [0-2]   100.0%      0.67
    Fixed, told the true rho            18   35.0      86.7% [80-91]     2.0%      0.00
    Pilot (5 tests), then fixed         17   16.9         0.0% [0-2]   100.0%      0.09
    Latent rho                          11   11.6         0.7% [0-4]    99.3%      0.19
    Latent rho and shape                23   40.0      64.0% [56-71]    18.0%      0.06


================================================================================
  Does inferring rho cost more than measuring it first?
================================================================================
  gap: latent costs -4.7 experiments against the pilot (< 1e-4), and recovers
     99.3% against 95.3% (exact McNemar 0.0703).
  geometric: latent costs -5.3 experiments against the pilot (< 1e-4), and recovers
     0.7% against 0.0% (exact McNemar 1.0000).

================================================================================
  What the latent arm pays for not being told rho
================================================================================
  gap: +0.4 experiments against an oracle that cannot exist in practice, 0.0006
  geometric: -23.4 experiments against an oracle that cannot exist in practice, < 1e-4
```

```
==========================================================================
  Recovering the assay model itself
==========================================================================
  gap: rho recovered to a median absolute error of 0.06, against a grid spacing of
     0.20-0.25 near the measured range; the 90% credible interval over grid cells
     covers the true value in 86.0% of runs.
     With the shape latent too, posterior mass on the true shape is 1.00 (median),
     above 0.9 in 100.0% of runs.
  geometric: rho recovered to a median absolute error of 0.19, against a grid spacing of
     0.20-0.25 near the measured range; the 90% credible interval over grid cells
     covers the true value in 36.0% of runs.
     With the shape latent too, posterior mass on the true shape is 1.00 (median),
     above 0.9 in 100.0% of runs.
```

References

[1] Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, Alex Y. Ge, Albert Xu, Marco Jost, Luke A. Gilbert, and Jonathan S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365(6455):786–793, 2019. doi: 10.1126/science.aax4438.

[2] Julia Joung, Sai Ma, Tristan Tay, et al. A transcription factor atlas of directed differentiation. *Cell*, 186 (1):209–229.e26, 2023. PMID 36608654.